

ARC COMPANION TACTICAL DEVICE

ARDUINO UNO R3 BUILD GUIDE

MILITARY TACTICAL GAMING DEVICE | 7-8 inch HDMI Screen | Arduino Uno R3 | RGB LED Underglow | Custom 3D Case

BUILD TIME: 1 WEEK | COST: 130 GBP | DIFFICULTY: BEGINNER-INTERMEDIATE

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PHASE 1: PROJECT OVERVIEW

What You Are Building:

A fully Arduino-controlled tactical gaming device with military aesthetic, LED underglow lighting, responsive touchscreen interface, blueprint tracking for ARC Raiders, raid timer with audio/visual feedback, and portable battery-powered operation.

Bill of Materials - Essential Components:

ITEM	QTY	COST GBP	SOURCE	ORDERED
7-inch HDMI LCD Touchscreen	1	55	Amazon UK	—
5V Power Bank 10000mAh	1	20	Amazon UK	—
WS2812B LED Strip 5m	1	15	Amazon	—
USB Hub (Powered)	1	8	Amazon UK	—
Black PLA Filament 500g	1	10	Amazon UK	—
Spray Paint Kit	1	15	Local	—
Hardware and Fasteners	1	5	Local	—
TOTAL		128	GBP	

What You Already Have (From Your Parts Stash):

Arduino Uno R3 (x3) | Jumper Wires | Resistors | Tactile Buttons | Breadboards | USB Cables | LEDs | Capacitors | QIDI Q2 3D Printer. Total value: 200+ GBP

PROJECT PLANNING NOTES:

SHOPPING CHECKLIST

Priority Order (Day 1):

PRIORITY	ITEM	COST	WHERE	RECEIVED
HIGH	7-inch HDMI LCD Touchscreen	55	Amazon Prime	—
HIGH	5V Power Bank 10000mAh	20	Amazon Prime	—
HIGH	WS2812B LED Strip	15	Amazon/Aliexpress	—
HIGH	USB Hub (Powered)	8	Amazon Prime	—
MEDIUM	Black PLA Filament	10	Amazon Prime	—
MEDIUM	Spray Paint Kit	15	Local Hardware	—
LOW	Mounting Hardware	5	Local Hardware	—

Budget Summary: Essential 98 GBP, Optional 30 GBP, You Have 200+ GBP. Total Project Cost: 128-130 GBP (versus 220 GBP for Raspberry Pi)

SHOPPING AND BUDGET NOTES:

PHASE 2: 3D PRINTING SETUP

Step 1: Prepare Design File

[Download OpenSCAD from openscad.org](#) | [Download case file: case_design_arduino.scad](#) | [Open in OpenSCAD](#) | [Press F5 to preview](#) | [Export to STL as arc_case.stl](#)

Step 2: QIDI Q2 Print Settings

SETTING	VALUE	CHECK
Nozzle Temperature	200 Celsius	■
Bed Temperature	60 Celsius (PLA)	■
Filament Type	Matte Black PLA	■
Layer Height	0.2mm	■
Infill Density	15 percent	■
Support Material	YES	■
Print Speed	50-60 mm per second	■
Estimated Time	12-14 hours	■

Step 3: Post-Print Processing (1 hour)

[Cool completely \(30 min\)](#) | [Remove from bed](#) | [Remove supports with pliers](#) | [Sand with 200-grit sandpaper \(5 min\)](#) | [Smooth edges with 400-grit \(10 min\)](#) | [Wipe clean with damp cloth](#) | [Let dry completely \(15 min\)](#)

3D PRINTING NOTES (Print issues, adjustments, etc):

PHASE 3: HARDWARE WIRING

Power Wiring Diagram

Power Bank (5V/2A) connects to USB Hub. Hub Port 1 to Arduino (Micro-B). Hub Port 2 to 7-inch LCD. LED Strip to 5V direct. Total draw: 700-1400 mA. Use 2A+ power supply minimum.

Button Wiring (4 Buttons - No Resistors Needed)

All buttons use internal Arduino pull-ups. Pin 2: MENU button to GND. Pin 3: SELECT button to GND. Pin 4: BACK button to GND. Pin 5: ACTION button to GND. Testing: Press each button - Serial Monitor should show activity.

LED Wiring (Status Indicators)

Pin 6 with 220-Ohm resistor to Red LED to GND. Pin 7 with 220-Ohm to Green LED to GND. Pin 8 with 220-Ohm to Blue LED to GND. Pin 9 with 220-Ohm to Yellow LED to GND. Pin 11 to Piezo Buzzer (optional) to GND. Resistors protect Arduino pins from LED current.

WIRING CHECKLIST AND NOTES:

PHASE 4: HARDWARE ASSEMBLY (2-3 hours)

Mount Arduino (15 min) | Wire Buttons (20 min) | Install LEDs (15 min) | Power Distribution (15 min) | Cable Management (10 min)

1. MOUNT ARDUINO: Place on breadboard for stability. Use double-sided foam tape to secure. Ensure all pins accessible. Check no cables blocking ports.
2. WIRE BUTTONS: Connect all 4 buttons to GND line. Connect button pins 2,3,4,5. Test each button. Use breadboard for clean connections.
3. INSTALL LEDs: Solder 220-Ohm resistors to each LED cathode. Connect anodes to Arduino pins 6,7,8,9. Connect all cathodes to common GND. Test that LEDs light.
4. POWER DISTRIBUTION: Connect USB hub to power bank. Connect Arduino USB to hub. Connect LCD USB to hub. Connect LED strip 5V to power bank. Test voltage with multimeter.
5. CABLE MANAGEMENT: Use zip ties to bundle cables. Keep power and signal cables separate. Route cables through case channels. Label cables for identification.

ASSEMBLY PROGRESS AND NOTES:

PHASE 5: ARDUINO CODE AND TESTING

Step 1: Install Arduino IDE (5 min)

Download from [arduino.cc](https://www.arduino.cc) | Install on your computer | Launch Arduino IDE | Tools > Board > Select Arduino Uno
| Tools > Port > Select USB COM port

Step 2: Upload Code (10 min)

Open `ARC_Companion_Arduino.ino` | Copy code into IDE | Click Verify button | Plug Arduino into computer |
Click Upload button | Should see Done uploading message

Step 3: Serial Monitor Test (5 min)

Tools > Serial Monitor | Set baud rate to 9600 | Should see ARC COMPANION message | Press MENU button -
Serial should show Menu | Press SELECT - should show Select | LEDs should flash on startup

CODE AND TESTING NOTES:

PHASE 6: CASE FINISHING

Case Painting (3-4 hours plus 24-hour dry time)

Mask screen opening and all ports with painter tape | Light sand with 400-grit | Apply 2 thin coats MATTE BLACK (5 min between) | Apply METALLIC GRAY accents | Apply 1 coat CLEAR POLYURETHANE | DRY 24 HOURS before assembly

Final Assembly (1 hour)

Verify all components working | Power on: All LEDs light | Press buttons: Serial responds | Test timer | Mount Arduino to case bottom | Install LED underglow strip | Mount LCD screen in bezel | Route all cables | Install acrylic diffuser | Close case carefully | Install 4 rubber feet | Final power test

FINISHING NOTES:

FINAL TESTING AND COMPLETION

TEST	EXPECTED RESULT	PASS	FAIL	NOTES
Power On	Arduino blue LED lights	■	■	
USB Hub	Hub LED lights	■	■	
7-inch LCD	Screen boots	■	■	
All Buttons	Serial responses correct	■	■	
All LEDs	All illuminate correctly	■	■	
LED Strip	Underglow lights	■	■	
Blueprint Search	Finds items correctly	■	■	
Timer	Counts seconds	■	■	
Overall Device	All systems operational	■	■	

BUILD COMPLETION NOTES:

PROJECT COMPLETE! You have successfully built a custom ARC Companion tactical device. Enjoy your handmade gaming device!